

# Bug Fix Impact Analysis

## METIMMOX-1 Economic Evaluation

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## 1 Summary

This report compares cost-effectiveness analysis results before and after addressing GitHub issue #141. The comparison assesses whether bug fixes or model changes affected results.

### Current Comparison:

- **Issue:** #141 - Red Team Code Review
- **Baseline:** Commit 9cdec7d (2026-02-19 10:55:10)

- **Fixed:** Commit cd6c906 (2026-02-19 14:10:02)
- **Models included:** Model A: Joint, Model B: Focused, Model C: Separate

## 2 Metadata Comparison

Table 1: Metadata Comparison

| Item                 | Baseline                          | Fixed                             |
|----------------------|-----------------------------------|-----------------------------------|
| Git Commit           | 9cdec7d                           | cd6c906                           |
| Timestamp            | 2026-02-19 10:55:10               | 2026-02-19 14:10:02               |
| R Version            | R version 4.3.2 (2023-10-31 ucrt) | R version 4.3.2 (2023-10-31 ucrt) |
| Issue Number         | 141                               | 141                               |
| Model Structure      | 2 (separate)                      | 2 (separate)                      |
| Multi-Model Snapshot | Yes                               | Yes                               |

### 2.1 Global Parameters

Table 2: Global Parameter Comparison

|                 | Parameter       | Baseline   | Fixed      | Changed |
|-----------------|-----------------|------------|------------|---------|
| WTP             | WTP             | 51,000     | 51,000     | No      |
| time_horizon    | time_horizon    | 520        | 520        | No      |
| n_samples       | n_samples       | 5,000      | 5,000      | No      |
| n_sim           | n_sim           | 5,000      | 5,000      | No      |
| cl              | cl              | 0.01923077 | 0.01923077 | No      |
| dr              | dr              | 0.04       | 0.04       | No      |
| DSA_mult        | DSA_mult        | 0.2        | 0.2        | No      |
| USE_BOTH_MODELS | USE_BOTH_MODELS | 0          | 0          | No      |
| MODEL_STRUCTURE | MODEL_STRUCTURE | 2          | 2          | No      |

### 3 Per-Model Comparisons

#### 3.1 Model A: Joint

##### 3.1.1 Base Case Results

Table 3: Model A: Joint - Incremental Cost Comparison

| Strategy | Baseline (EUR) | Fixed (EUR) | Difference (EUR) |
|----------|----------------|-------------|------------------|
| control  | 0.00           | 0.00        | 0.00             |
| crp      | 34,003.49      | 34,003.49   | 0.00             |
| tlr      | 64,079.67      | 64,079.67   | 0.00             |
| tmb_braf | 42,665.89      | 42,665.89   | 0.00             |

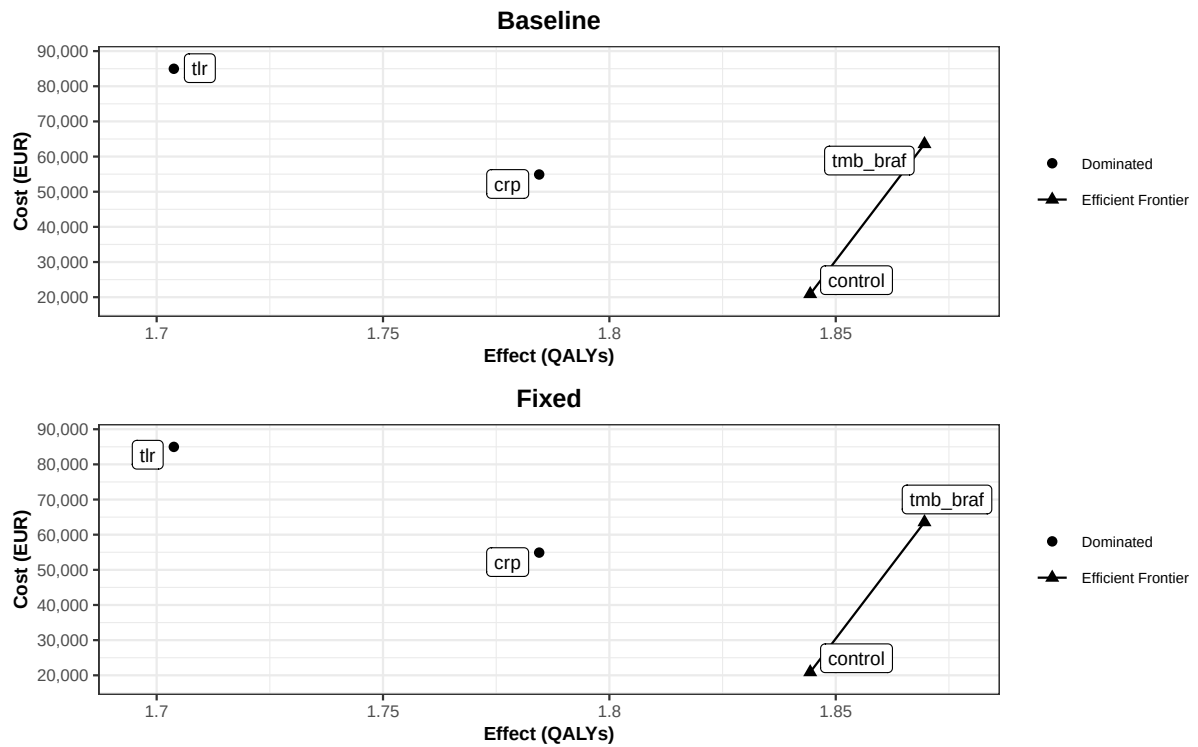
Table 4: Model A: Joint - Incremental Effect Comparison

| Strategy | Baseline (QALYs) | Fixed (QALYs) | Difference (QALYs) |
|----------|------------------|---------------|--------------------|
| control  | 0.0000           | 0.0000        | 0.0000             |
| crp      | -0.0599          | -0.0599       | 0.0000             |
| tlr      | -0.1406          | -0.1406       | 0.0000             |
| tmb_braf | 0.0252           | 0.0252        | 0.0000             |

Table 5: Model A: Joint - ICER Comparison

| Strategy | Baseline (EUR/QALY) | Fixed (EUR/QALY) | Difference (EUR/QALY) |
|----------|---------------------|------------------|-----------------------|
| control  | —                   | —                | —                     |
| crp      | —                   | —                | —                     |
| tlr      | —                   | —                | —                     |
| tmb_braf | 1,690,480           | 1,690,480        | 0                     |

### 3.1.2 Efficiency Frontier



TableGrob (2 x 1) “arrange”: 2 grobs z cells name grob 1 1 (1-1,1-1) arrange gtable[layout] 2 2 (2-2,1-1) arrange gtable[layout]

### 3.1.3 Net Monetary Benefit

Table 6: Model A: Joint - NMB at WTP = EUR 51,000

|   | Strategy | Baseline (EUR)   | Fixed (EUR)      | Difference (EUR) |
|---|----------|------------------|------------------|------------------|
| 1 | control  | <b>73,168.05</b> | <b>73,168.05</b> | <b>0.00</b>      |
| 2 | crp      | 36,111.38        | 36,111.38        | 0.00             |
| 4 | tmb_braf | 31,789.34        | 31,789.34        | 0.00             |
| 3 | tlr      | 1,919.73         | 1,919.73         | 0.00             |

**Optimal Strategy:** Unchanged (control)

### 3.1.4 PSA Results

Table 7: Model A: Joint - PSA Cost Results

| Strategy | Baseline Mean (2.5%; 97.5%)                   | Fixed Mean (2.5%; 97.5%)                      | Difference (EUR) |
|----------|-----------------------------------------------|-----------------------------------------------|------------------|
| control  | EUR 21,168.23 (EUR 16,131.56; EUR 27,280.45)  | EUR 21,169.03 (EUR 15,999.01; EUR 27,222.90)  | 0.81             |
| crp      | EUR 54,946.66 (EUR 39,037.76; EUR 75,205.34)  | EUR 54,806.03 (EUR 38,836.45; EUR 74,193.07)  | -140.63          |
| tlr      | EUR 85,007.04 (EUR 55,739.24; EUR 121,997.19) | EUR 84,998.34 (EUR 56,148.61; EUR 120,638.87) | -8.70            |
| tmb_braf | EUR 63,486.53 (EUR 44,829.42; EUR 87,414.23)  | EUR 63,212.14 (EUR 44,532.94; EUR 86,258.98)  | -274.39          |

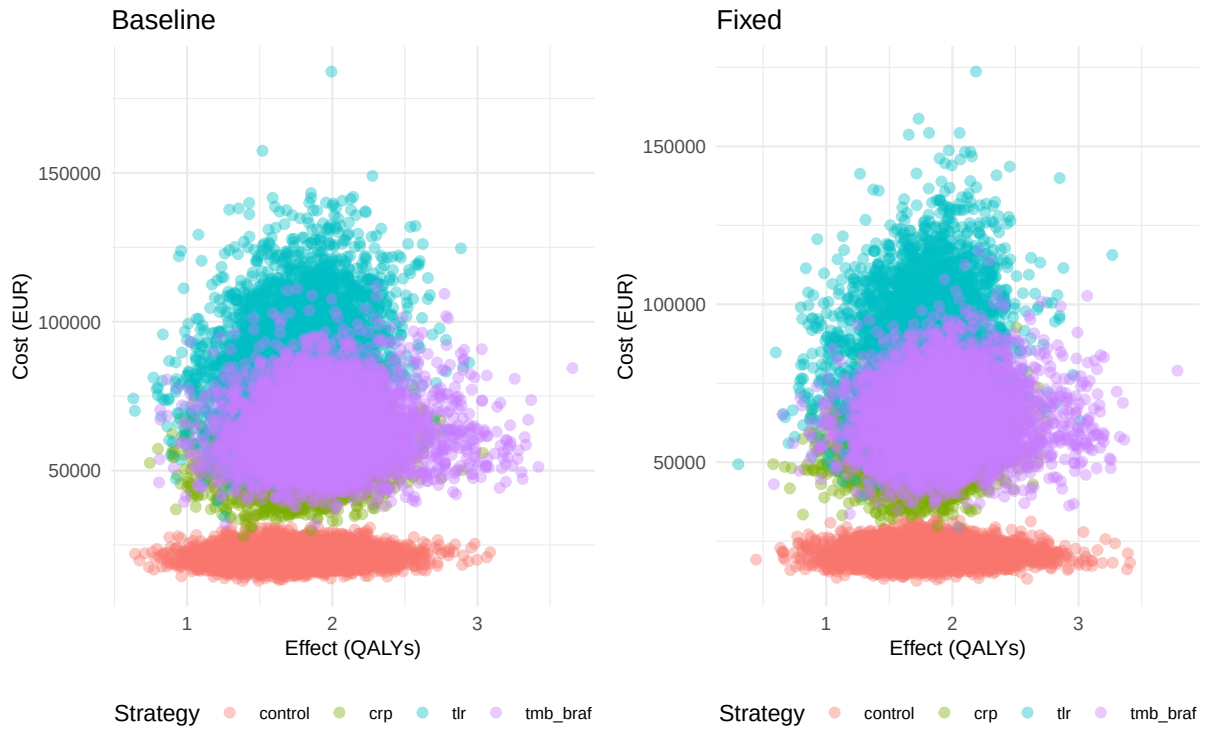
Table 8: Model A: Joint - PSA Effect Results

| Strategy | Baseline Mean (2.5%; 97.5%) | Fixed Mean (2.5%; 97.5%) | Difference (QALYs) |
|----------|-----------------------------|--------------------------|--------------------|
| control  | 1.7058 (1.0948; 2.3939)     | 1.7829 (1.0989; 2.5700)  | 0.0772             |
| crp      | 1.8134 (1.2485; 2.3111)     | 1.8111 (1.2010; 2.3319)  | -0.0023            |
| tlr      | 1.7539 (1.1406; 2.3527)     | 1.7566 (1.1112; 2.3683)  | 0.0027             |
| tmb_braf | 1.9349 (1.2726; 2.7834)     | 1.9338 (1.2636; 2.7854)  | -0.0011            |

Table 9: Model A: Joint - PSA ICER Results

| Strategy | Baseline Mean ICER | Fixed Mean ICER    |
|----------|--------------------|--------------------|
| control  | —                  | —                  |
| crp      | EUR 313,851/QALY   | EUR 1,195,364/QALY |
| tlr      | EUR 1,326,685/QALY | —                  |
| tmb_braf | EUR 184,704/QALY   | EUR 278,717/QALY   |

### 3.1.5 PSA Scatter Plots



TableGrob (1 x 2) “arrange”: 2 grobs z cells name grob 1 1 (1-1,1-1) arrange gtable[layout] 2 2 (1-1,2-2) arrange gtable[layout]

## 3.2 Model B: Focused

### 3.2.1 Base Case Results

Table 10: Model B: Focused - Incremental Cost Comparison

| Strategy | Baseline (EUR) | Fixed (EUR) | Difference (EUR) |
|----------|----------------|-------------|------------------|
| control  | 0.00           | 0.00        | 0.00             |
| crp      | 33,614.22      | 33,857.23   | 243.00           |
| tlr      | 64,906.87      | 65,149.87   | 243.00           |
| tmb_braf | 40,038.44      | 40,281.45   | 243.00           |

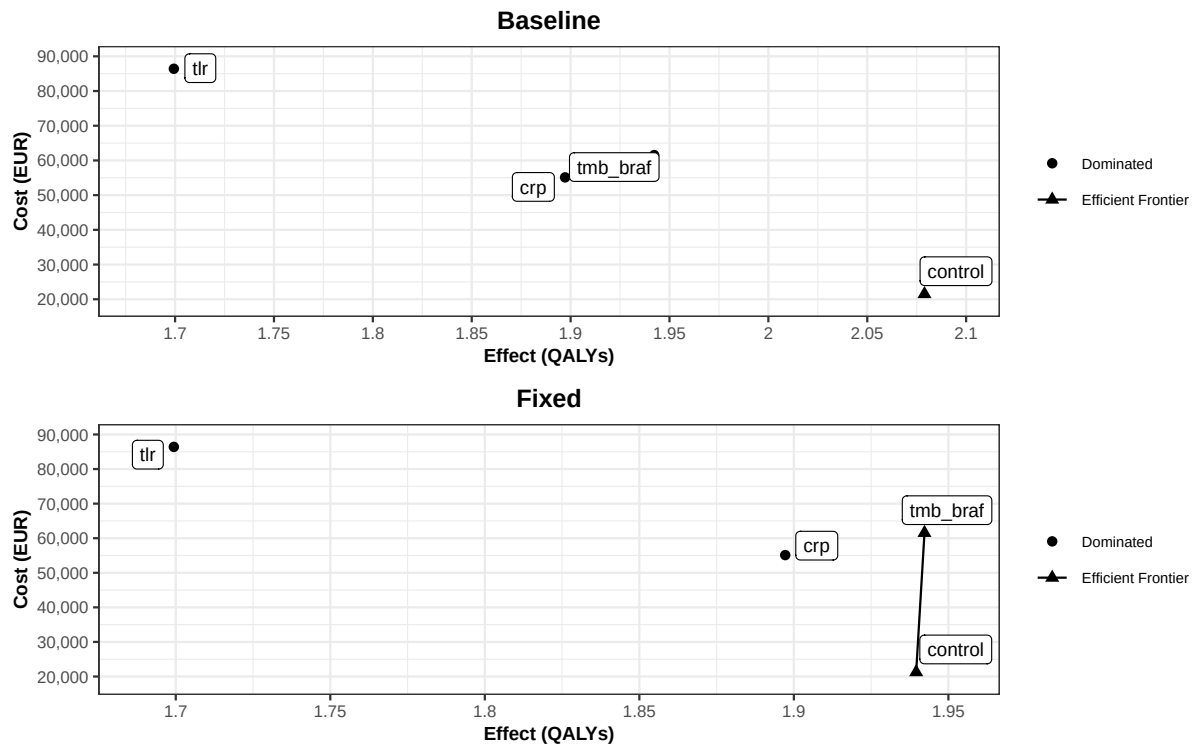
Table 11: Model B: Focused - Incremental Effect Comparison

| Strategy | Baseline (QALYs) | Fixed (QALYs) | Difference (QALYs) |
|----------|------------------|---------------|--------------------|
| control  | 0.0000           | 0.0000        | 0.0000             |
| crp      | -0.1817          | -0.0424       | 0.1394             |
| tlr      | -0.3796          | -0.2402       | 0.1394             |
| tmb_braf | -0.1367          | 0.0027        | 0.1394             |

Table 12: Model B: Focused - ICER Comparison

| Strategy | Baseline (EUR/QALY) | Fixed (EUR/QALY) | Difference (EUR/QALY) |
|----------|---------------------|------------------|-----------------------|
| control  | —                   | —                | —                     |
| crp      | —                   | —                | —                     |
| tlr      | —                   | —                | —                     |
| tmb_braf | —                   | 14,980,066       | —                     |

### 3.2.2 Efficiency Frontier



TableGrob (2 x 1) “arrange”: 2 grobs z cells name grob 1 1 (1-1,1-1) arrange gtable[layout] 2 2 (2-2,1-1) arrange gtable[layout]

### 3.2.3 Net Monetary Benefit

Table 13: Model B: Focused - NMB at WTP = EUR 51,000

|   | Strategy       | Baseline (EUR)   | Fixed (EUR)      | Difference (EUR) |
|---|----------------|------------------|------------------|------------------|
| 1 | <b>control</b> | <b>84,521.48</b> | <b>77,657.39</b> | <b>-6,864.09</b> |
| 2 | crp            | 41,639.57        | 41,639.57        | 0.00             |
| 4 | tmb_braf       | 37,513.08        | 37,513.08        | 0.00             |
| 3 | tlr            | 257.20           | 257.20           | 0.00             |

**Optimal Strategy:** Unchanged (control)

### 3.3 Model C: Separate

#### 3.3.1 Base Case Results

Table 14: Model C: Separate - Incremental Cost Comparison

| Strategy | Baseline (EUR) | Fixed (EUR) | Difference (EUR) |
|----------|----------------|-------------|------------------|
| control  | 0.00           | 0.00        | 0.00             |
| crp      | 34,068.60      | 34,292.03   | 223.42           |
| tlr      | 66,766.64      | 66,990.06   | 223.42           |
| tmb_braf | 46,441.58      | 46,665.00   | 223.42           |

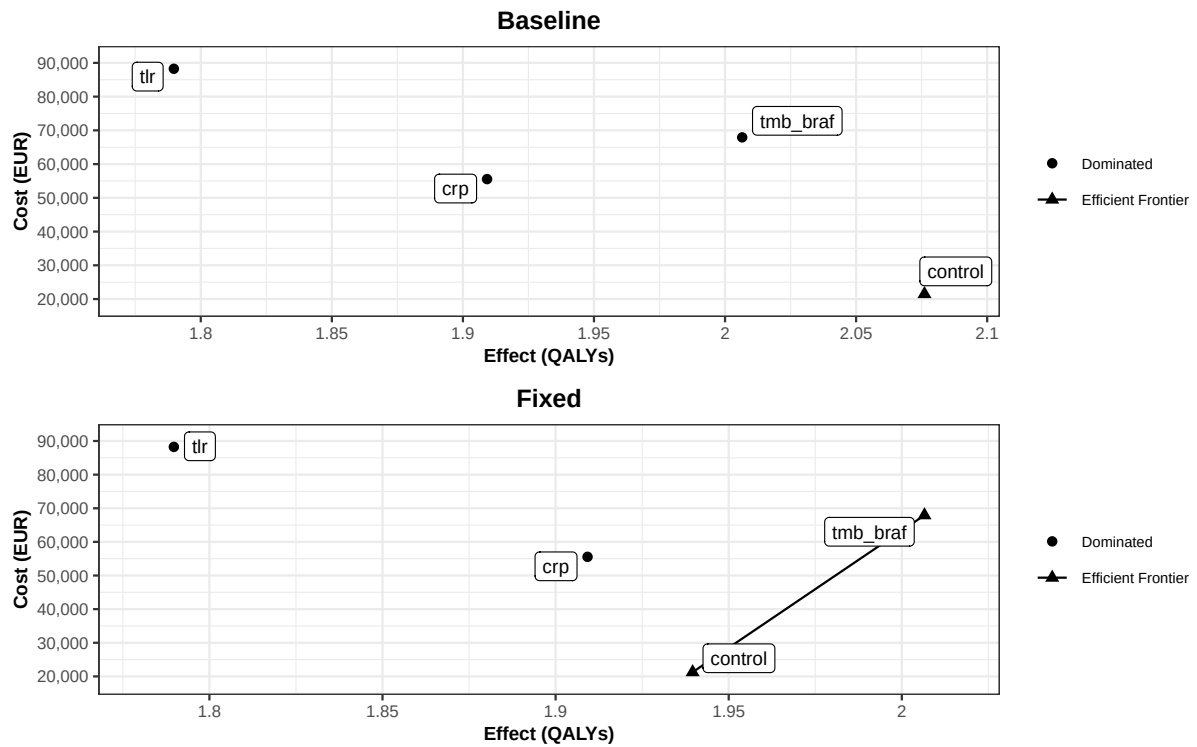
Table 15: Model C: Separate - Incremental Effect Comparison

| Strategy | Baseline (QALYs) | Fixed (QALYs) | Difference (QALYs) |
|----------|------------------|---------------|--------------------|
| control  | 0.0000           | 0.0000        | 0.0000             |
| crp      | -0.1669          | -0.0304       | 0.1365             |
| tlr      | -0.2864          | -0.1498       | 0.1365             |
| tmb_braf | -0.0695          | 0.0670        | 0.1365             |

Table 16: Model C: Separate - ICER Comparison

| Strategy | Baseline (EUR/QALY) | Fixed (EUR/QALY) | Difference (EUR/QALY) |
|----------|---------------------|------------------|-----------------------|
| control  | —                   | —                | —                     |
| crp      | —                   | —                | —                     |
| tlr      | —                   | —                | —                     |
| tmb_braf | —                   | 696,842          | —                     |

### 3.3.2 Efficiency Frontier



TableGrob (2 x 1) “arrange”: 2 grobs z cells name grob 1 1 (1-1,1-1) arrange gtable[layout] 2 2 (2-2,1-1) arrange gtable[layout]

### 3.3.3 Net Monetary Benefit

Table 17: Model C: Separate - NMB at WTP = EUR 51,000

|   | Strategy       | Baseline (EUR)   | Fixed (EUR)      | Difference (EUR) |
|---|----------------|------------------|------------------|------------------|
| 1 | <b>control</b> | <b>84,395.88</b> | <b>77,657.39</b> | <b>-6,738.50</b> |
| 2 | crp            | 41,817.15        | 41,817.15        | 0.00             |
| 4 | tmb_braf       | 34,407.67        | 34,407.67        | 0.00             |
| 3 | tlr            | 3,025.08         | 3,025.08         | 0.00             |

**Optimal Strategy:** Unchanged (control)

## 4 Cross-Model Summary

Table 18: Cross-Model Impact Summary

|          | Model             | Opt. (BL) | Opt. (Fix) | Changed? | Max Cost Delta (EUR) | Max Effect Delta (QALYs) | Max ICER Delta (EUR/QALY) |
|----------|-------------------|-----------|------------|----------|----------------------|--------------------------|---------------------------|
| joint    | Model A: Joint    | control   | control    | No       | 0.00                 | 0.0000                   | 0                         |
| focused  | Model B: Focused  | control   | control    | No       | 243.00               | 0.1394                   | -Inf                      |
| separate | Model C: Separate | control   | control    | No       | 223.42               | 0.1365                   | -Inf                      |



## 5 Comparison Across Recent Issues

### 5.1 Available Snapshot Pairs

Table 19: All Available Snapshot Pairs

| Issue    |            | Title                                                                                        | Baseline Commit | Fixed Commit   |
|----------|------------|----------------------------------------------------------------------------------------------|-----------------|----------------|
| <b>2</b> | <b>141</b> | <b>Red Team Code Review</b>                                                                  | <b>9cdec7d</b>  | <b>cd6c906</b> |
| 1        | 118        | [P2] change model B                                                                          | 4e55627         | 10df9fe        |
| 3        | 18         | [P1] other costs                                                                             | 71f8230         | 71f8230        |
| 19       | 99         | [P2] Control model formula inconsistency between base case and PSA                           | 376b4c6         | d1eef51        |
| 18       | 97         | [P1] Inconsistent population for averaging in PSA (control vs biomarker strategies)          | 457533e         | 457533e        |
| 6        | 47         | [P2] Missing input parameter validation in model_fun()                                       | 2411bdb         | 2411bdb        |
| 5        | 46         | [P1] Bootstrap sample index not validated in PSA mode                                        | e8b3c27         | 2d5181c        |
| 17       | 94         | [P2] PSA errors                                                                              | e77415f         | 3f98095        |
| 12       | 75         | [P0] Biomarker Prevalence Not Treated as Uncertain Parameter                                 | 2b04ea7         | c15aa47        |
| 16       | 90         | Sensitivity analysis: independent biomarker-specific survival models                         | 9e1b590         | 7d68bca        |
| 13       | 76         | [P0] PFS Can Exceed OS (Biological Impossibility Not Enforced)                               | 64cf543         | 64cf543        |
| 15       | 87         | [P0] Fix PSA methodology: Replace single-patient sampling with subgroup population averaging | 42d4acb         | 0a15902        |
| 14       | 86         | [P1] Mltpl documentation issues                                                              | ee96c31         | f3e5604        |
| 11       | 70         | PSA uses inconsistent                                                                        | 998dacd         | 605ff90        |

## 5.2 Multi-Issue Impact Summary

This table summarizes the impact of bug fixes across the most recent issues:

Table 20: Summary of Bug Fix Impacts (5 Most Recent Issues)

| Issue      | Title                                                                                     | Opt. (BL)      | Opt. (Fix)     | Chg?      | Max Cost<br>Delta (EUR) | Max Effect<br>Delta<br>(QALYs) | Max ICER<br>Delta<br>(EUR/QALY) |
|------------|-------------------------------------------------------------------------------------------|----------------|----------------|-----------|-------------------------|--------------------------------|---------------------------------|
| <b>141</b> | <b>Red Team Code Review</b>                                                               | <b>control</b> | <b>control</b> | <b>No</b> | <b>0.00</b>             | <b>0.0000</b>                  | <b>0</b>                        |
| 118        | [P2] change model B                                                                       | control        | control        | No        | 0.00                    | 0.0000                         | 0                               |
| 18         | [P1] other costs                                                                          | control        | control        | No        | 1,314.18                | 0.0000                         | 44,593                          |
| 99         | [P2] Control model formula<br>inconsistency between base<br>case and PSA                  | control        | control        | No        | 0.00                    | 0.0000                         | 0                               |
| 97         | [P1] Inconsistent population<br>for averaging in PSA (control<br>vs biomarker strategies) | control        | control        | No        | 0.00                    | 0.0000                         | 0                               |

## 6 Summary

The comparison for issue #141 shows:

- **Models compared:** Model A: Joint, Model B: Focused, Model C: Separate
- **Optimal Strategy (Model A: Joint):** Unchanged (control)
- **Maximum Cost Change:** EUR 0.00 (minor)
- **Maximum Effect Change:** 0.000 QALYs (minor)
- **Maximum ICER Change:** EUR 0/QALY (minor)
- **Parameter Changes:** None detected

**Interpretation:** Changes are within expected bounds for bug fixes that improve numerical stability or code quality.

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**Report completed on:** 2026-02-20

**Repository:** ben-geisler/METIMMOX-1

**Report version:** 3.0